

43.1 Kernel and local polynomial regression

Drop the linear predictor. Keep one covariate, keep additive errors, and assume nothing about the shape of the regression function except that it is smooth:

$$y_i = f(x_i) + \varepsilon_i, \quad E(\varepsilon_i) = 0, \quad \text{Var}(\varepsilon_i) = \sigma^2, \quad i = 1, \dots, n, \quad (43.1.1)$$

with independent errors and design points $x_1, \dots, x_n$ in an interval $[a, b]$. The unknown is now a *function*, not a vector of coefficients, and there is no finite-dimensional parameter to estimate consistently. What replaces it is an assumption of *local* regularity: f does not change much over a short interval, so observations with covariate values near x carry information about $f(x)$.

That sentence is the whole idea of this section. Fix a target point x , take the observations whose covariates lie in a window around it, fit something simple to them by least squares, and read off the fitted value at x ; move x across the interval and the fitted values trace a curve. Everything else is bookkeeping: which observations count as “near”, how much they count, and what “something simple” means.

43.1.1 Local averages

Example. (The running mean)

The crudest smoother averages the responses in a window: fix a half-width h and set $\hat{f}(x) = |N(x)|^{-1} \sum_{i \in N(x)} y_i$ with $N(x) = \{i : |x_i - x| \leq h\}$. Three things go wrong at once. The curve is a step function, since $\hat{f}$ changes only when a point enters or leaves the window; every point in the window counts the same, at the centre or at the edge; and near $x = a$ the window is one-sided, so the average estimates f somewhere to the right of a .

Weighting repairs the first two defects: count a point by a weight that decreases smoothly with distance and vanishes outside the window. The third is deeper, and is repaired by fitting a line rather than a constant. Both repairs fit in one definition.

Definition 43.1.1 — Kernel weights and the local polynomial estimator

A **kernel** is a bounded function K with $\int K(u) du = 1$ and $K(u) = K(-u)$, here always nonnegative and supported on $[-1, 1]$. Its moments are $\mu_j = \int u^j K(u) du$ and $\nu_j = \int u^j K(u)^2 du$, with $R(K) = \nu_0$. For a **bandwidth** $h > 0$ write $K_h(u) = h^{-1} K(u/h)$.

Fix a degree $q \geq 0$. The **local polynomial estimator of degree q** at the point x is $\hat{f}(x) = \hat{b}_0$, where $(\hat{b}_0, \dots, \hat{b}_q)$ minimizes

$$\sum_{i=1}^n K_h(x_i - x) \left\{ y_i - \sum_{j=0}^q b_j (x_i - x)^j \right\}^2. \quad (43.1.2)$$

For $q = 0$ this is the **local constant** or **Nadaraya–Watson** estimator

$$\hat{f}(x) = \frac{\sum_i K_h(x_i - x) y_i}{\sum_i K_h(x_i - x)}, \quad (43.1.3)$$

and for $q = 1$ the **local linear** estimator.

Three kernels are used throughout: the uniform kernel $K(u) = \frac{1}{2}$ on $[-1, 1]$, which gives the running mean back; the **Epanechnikov** kernel $K(u) = \frac{3}{4}(1 - u^2)$ on $[-1, 1]$, with $\mu_2 = 1/5$ and $R(K) = 3/5$; and the standard normal density, whose unbounded support does not literally satisfy the conditions below, though the conclusions are the same. The kernel matters far less than h , as [Exercise \(B2\)](#) quantifies.

[Eq. 43.1.2](#) is a weighted least squares problem (Theorem 21.3.1) with model matrix and weight matrix

$$\mathbf{U}_x = \begin{pmatrix} 1 & x_1 - x & \cdots & (x_1 - x)^q \\ \vdots & \vdots & & \vdots \\ 1 & x_n - x & \cdots & (x_n - x)^q \end{pmatrix}, \quad \mathbf{W}_x = \text{diag}\{K_h(x_i - x)\}_{i=1}^n.$$

Nothing in it is new; what is new is that it is solved afresh at every x , and that the weights, not a global model, decide which observations matter.

43.1.2 The estimator is linear, and it reproduces polynomials

Proposition 43.1.1 — Equivalent kernel weights

Suppose $\mathbf{U}_x^\top \mathbf{W}_x \mathbf{U}_x$ is nonsingular. Then $\hat{f}(x) = \sum_i \ell_i(x) y_i$ with

$$\ell_i(x) = \mathbf{e}_1^\top (\mathbf{U}_x^\top \mathbf{W}_x \mathbf{U}_x)^{-1} \mathbf{U}_x^\top \mathbf{W}_x \mathbf{e}_i, \quad (43.1.4)$$

where $\mathbf{e}_1$ is the first standard basis vector, and

- (a) $\sum_i \ell_i(x) (x_i - x)^j = \delta_{j0}$ for $j = 0, \dots, q$, where δ_{j0} is one if $j = 0$ and zero otherwise;
- (b) consequently $\hat{f}$ reproduces polynomials: if f is a polynomial of degree at most q then $\sum_i \ell_i(x) f(x_i) = f(x)$ for every x , whatever the design, the kernel and the bandwidth;
- (c) the weights do not depend on $\mathbf{y}$, so $\text{Var}\{\hat{f}(x)\} = \sigma^2 \sum_i \ell_i(x)^2$.

Proof. The weighted normal equations (Theorem 6.4.2) are $\mathbf{U}_x^\top \mathbf{W}_x \mathbf{U}_x \hat{\mathbf{b}} = \mathbf{U}_x^\top \mathbf{W}_x \mathbf{y}$, and $\hat{f}(x) = \mathbf{e}_1^\top \hat{\mathbf{b}}$, which is [Eq. 43.1.4](#). For (a), the j th column of $\mathbf{U}_x$ has entries $(x_i - x)^j$, so $\sum_i \ell_i(x) (x_i - x)^j = \mathbf{e}_1^\top (\mathbf{U}_x^\top \mathbf{W}_x \mathbf{U}_x)^{-1} \mathbf{U}_x^\top \mathbf{W}_x \mathbf{U}_x \mathbf{e}_{j+1} = \mathbf{e}_1^\top \mathbf{e}_{j+1}$. Part (b) follows by writing $f(x_i) = \sum_{j \leq q} c_j (x_i - x)^j$ with $c_0 = f(x)$, and (c) because $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$. ■

Part (a) is the engine of everything that follows. The weights sum to one, so the estimator has no bias against a constant; for $q \geq 1$ they also annihilate $(x_i - x)$, so it has no bias against a

linear trend either. That is exactly what the running mean lacked.

The weight profile $u \mapsto \ell_i(x)$, with $u = (x_i - x)/h$, is the **equivalent kernel** of the estimator. For $q = 1$ at an interior point it is asymptotically $K(u)/\{nhg(x)\}$: a local linear fit *is* a kernel average, with the kernel unchanged. For $q = 3$ it is $(\mu_4 - \mu_2 u^2)K(u)/(\mu_4 - \mu_2^2)$, which takes negative values — the price of killing the h^2 bias term.

43.1.3 Bias and variance

Four assumptions make the asymptotics exact rather than heuristic. A random design would work too, at the cost of replacing every statement by one that holds in probability; the proof below is for a *regular* design, where the argument is a clean Riemann-sum estimate.

(A) Design. $x_i = G^{-1}\{(i - \frac{1}{2})/n\}$ for $i = 1, \dots, n$, where G is a distribution function on $[a, b]$ whose density g is continuously differentiable and bounded away from zero.

(B) Kernel. K is as in Definition 43.1.1, of bounded variation, with $\mu_2 > 0$.

(C) Function. f is twice continuously differentiable on $[a, b]$.

(D) Bandwidth. $h = h_n \rightarrow 0$ and $nh \rightarrow \infty$.

Assumption (A) holds exactly for an equally spaced or quantile-spaced design. The next lemma, the only analysis in this chapter, says that a kernel-weighted sum over such a design is an integral plus a bounded error.

Lemma 43.1.2 — Kernel sums are integrals

Let φ be of bounded variation on $\mathbb{R}$ with support in $[-1, 1]$ and total variation $V(\varphi)$. Under (A), for every x and every $h > 0$,

$$\left| \sum_{i=1}^n \varphi\left(\frac{x_i - x}{h}\right) - nh \int_{(a-x)/h}^{(b-x)/h} \varphi(u) g(x + hu) du \right| \leq V(\varphi). \quad (43.1.5)$$

Proof. Put $\Psi(v) = \varphi\{(G^{-1}(v) - x)/h\}$ for $v \in [0, 1]$. Since G^{-1} is increasing, Ψ has the same total variation as $t \mapsto \varphi\{(t - x)/h\}$, namely $V(\varphi)$. The sum on the left is $\sum_i \Psi\{(i - \frac{1}{2})/n\}$, a midpoint sum. Write I_i for the i th subinterval of $[0, 1]$ of length $1/n$ and V_i for the variation of Ψ on I_i . For $v \in I_i$, $|\Psi\{(i - \frac{1}{2})/n\} - \Psi(v)| \leq V_i$, so $|n^{-1}\Psi\{(i - \frac{1}{2})/n\} - \int_{I_i} \Psi| \leq V_i/n$. Summing over i and using $\sum_i V_i = V(\varphi)$ gives $|\sum_i \Psi\{(i - \frac{1}{2})/n\} - n \int_0^1 \Psi| \leq V(\varphi)$. Finally substitute $t = G^{-1}(v)$, so that $dv = g(t) dt$, and then $t = x + hu$. ■

Applied with $\varphi(u) = K(u)u^j$ and with $\varphi = K^2 u^j$, the lemma turns every sum that appears below into a moment of the kernel. Write

$$S_{n,j} = \sum_{i=1}^n K_h(x_i - x)(x_i - x)^j,$$

and let $\mathbf{A} = (\mu_{j+k})_{j,k=0}^q$ and $\mathbf{A}^* = (\nu_{j+k})_{j,k=0}^q$ be the kernel's moment matrices, which are

positive definite because μ_{j+k} and ν_{j+k} are the Gram matrices of $1, u, \dots, u^q$ in $L^2(K du)$ and $L^2(K^2 du)$.

Theorem 43.1.3 — Bias and variance of a local polynomial estimate

Assume (A)–(D) and let x satisfy $[x - h, x + h] \subset (a, b)$.

(a) (*Local linear*, $q = 1$.)

$$\mathbb{E}\{\hat{f}(x)\} - f(x) = \frac{1}{2}\mu_2 f''(x)h^2 + o(h^2), \quad \text{Var}\{\hat{f}(x)\} = \frac{\sigma^2 R(K)}{nh g(x)}\{1 + o(1)\}.$$

(b) (*Local constant*, $q = 0$.) The variance is the same to first order, and

$$\mathbb{E}\{\hat{f}(x)\} - f(x) = \mu_2 h^2 \left\{ \frac{1}{2}f''(x) + \frac{f'(x)g'(x)}{g(x)} \right\} + o(h^2). \quad (43.1.6)$$

(c) (*At the boundary*.) Let $x = a + ch$ with $c \in [0, 1)$ fixed, and write $\mu_{j,c} = \int_{-c}^1 u^j K(u) du$. Then the local constant estimator has bias $f'(a) h \mu_{1,c}/\mu_{0,c} + O(h^2)$, of exact order h unless $f'(a) = 0$, while the local linear estimator still has bias $O(h^2)$.

(d) (*Optimal bandwidth*.) For the local linear estimator with $f''(x) \neq 0$, the pointwise mean squared error $\frac{1}{4}\mu_2^2 f''(x)^2 h^4 + \sigma^2 R(K)/\{nhg(x)\}$ is minimized at

$$h^*(x) = \left\{ \frac{\sigma^2 R(K)}{g(x) \mu_2^2 f''(x)^2} \right\}^{1/5} n^{-1/5}, \quad (43.1.7)$$

and the minimum value is $\frac{5}{4}\{\mu_2^2 f''(x)^2\}^{1/5}\{\sigma^2 R(K)/g(x)\}^{4/5} n^{-4/5}$.

Proof. Write $u_i = (x_i - x)/h$, let $\mathbf{U}$ have rows $(1, u_i, \dots, u_i^q)$ and $\mathbf{W} = \mathbf{W}_x$. Because $\mathbf{U}_x = \mathbf{U} \text{diag}(1, h, \dots, h^q)$ and the first diagonal entry is one, Eq. 43.1.4 gives $\ell_i(x) = \mathbf{e}_1^\top (\mathbf{U}^\top \mathbf{W} \mathbf{U})^{-1} \mathbf{U}^\top \mathbf{W} \mathbf{e}_i$.

Moment sums. By Lemma 43.1.2 with $\varphi(u) = K(u)u^j$, and because $[x - h, x + h] \subset (a, b)$ makes the range of integration all of $[-1, 1]$,

$$S_{n,j} = h^{j-1} \sum_i K(u_i) u_i^j = nh^j \left\{ \int K(u) u^j g(x + hu) du + O((nh)^{-1}) \right\}.$$

Since $g \in C^1$, $g(x + hu) = g(x) + hug'(x + \theta hu)$ with $\theta \in (0, 1)$, and g' is uniformly continuous, so

$$S_{n,j} = nh^j \{g(x)\mu_j + hg'(x)\mu_{j+1} + o(h) + O((nh)^{-1})\}. \quad (43.1.8)$$

Because $(\mathbf{U}^\top \mathbf{W} \mathbf{U})_{jk} = \sum_i K_h(x_i - x) u_i^{j+k} = h^{-(j+k)} S_{n,j+k}$, this says $\mathbf{U}^\top \mathbf{W} \mathbf{U} = n\{g(x)\mathbf{A} + O(h) + O((nh)^{-1})\}$. In the same way, with $\varphi = K^2 u^j$, $\mathbf{U}^\top \mathbf{W}^2 \mathbf{U} = (n/h)\{g(x)\mathbf{A}^* + O(h) + O((nh)^{-1})\}$. Assumption (D) makes both error terms vanish, so $\mathbf{U}^\top \mathbf{W} \mathbf{U}$ is eventually nonsingular and Proposition 43.1.1 applies.

Variance. By [Proposition 43.1.1\(c\)](#),

$$\text{Var}\{\hat{f}(x)\} = \sigma^2 \mathbf{e}_1^\top (\mathbf{U}^\top \mathbf{W} \mathbf{U})^{-1} (\mathbf{U}^\top \mathbf{W}^2 \mathbf{U}) (\mathbf{U}^\top \mathbf{W} \mathbf{U})^{-1} \mathbf{e}_1 = \frac{\sigma^2 \mathbf{e}_1^\top \mathbf{A}^{-1} \mathbf{A}^* \mathbf{A}^{-1} \mathbf{e}_1}{nh g(x)} \{1 + o(1)\}.$$

For $q = 0$, $\mathbf{A} = (1)$ and $\mathbf{A}^* = (R(K))$. For $q = 1$ the symmetry of K makes $\mu_1 = 0$, so $\mathbf{A} = \text{diag}(1, \mu_2)$, $\mathbf{A}^{-1} \mathbf{e}_1 = \mathbf{e}_1$ and the quadratic form is $\nu_0 = R(K)$. Both give the stated variance.

Bias, $q = 1$. By [Proposition 43.1.1\(a\)](#) the weights annihilate constants and linear terms, so with a second-order Taylor expansion of f about x , $f(x_i) = f(x) + f'(x)(x_i - x) + \frac{1}{2}f''(\xi_i)(x_i - x)^2$ for some ξ_i between x and x_i ,

$$\mathbb{E}\{\hat{f}(x)\} - f(x) = \frac{1}{2}f''(x) \sum_i \ell_i(x)(x_i - x)^2 + \frac{1}{2} \sum_i \ell_i(x)\{f''(\xi_i) - f''(x)\}(x_i - x)^2.$$

For the first sum, $\sum_i \ell_i(x)(x_i - x)^2 = h^2 \mathbf{e}_1^\top (\mathbf{U}^\top \mathbf{W} \mathbf{U})^{-1} \mathbf{U}^\top \mathbf{W} \mathbf{u}^{(2)}$ with $\mathbf{u}_i^{(2)} = u_i^2$, and $(\mathbf{U}^\top \mathbf{W} \mathbf{u}^{(2)})_j = \sum_i K_h(x_i - x) u_i^{j+2} = h^{-(j+2)} S_{n,j+2}$, which by [Eq. 43.1.8](#) equals $n\{g(x)\mu_{j+2} + O(h) + O((nh)^{-1})\}$. Hence

$$\sum_i \ell_i(x)(x_i - x)^2 = h^2 [\mathbf{e}_1^\top \mathbf{A}^{-1} (\mu_2, \mu_3)^\top + O(h) + O((nh)^{-1})] = h^2 \{\mu_2 + o(1)\},$$

using $\mu_3 = 0$. For the second sum, only $|x_i - x| \leq h$ contributes, so $|f''(\xi_i) - f''(x)| \leq \omega(h)$, the modulus of continuity of f'' , which tends to zero. Since $(\mathbf{U}^\top \mathbf{W} \mathbf{U})^{-1} = O(n^{-1})$ entrywise and $\sum_i K_h(x_i - x) |u_i|^j = O(n)$ by [Lemma 43.1.2](#), we get $\sum_i \ell_i(x) |(x_i - x)^2| = O(h^2)$, and the second sum is $O\{\omega(h)h^2\} = o(h^2)$. This proves (a).

Bias, $q = 0$. Now only constants are annihilated, so a term in $f'(x)$ survives:

$$\mathbb{E}\{\hat{f}(x)\} - f(x) = f'(x) \frac{S_{n,1}}{S_{n,0}} + \frac{1}{2}f''(x) \frac{S_{n,2}}{S_{n,0}} + o(h^2),$$

the remainder being controlled as above. By [Eq. 43.1.8](#) and $\mu_1 = 0$, $S_{n,0} = n\{g(x) + o(h)\}$ and $S_{n,1} = nh^2\{g'(x)\mu_2 + o(1)\}$, so $S_{n,1}/S_{n,0} = h^2\mu_2 g'(x)/g(x) + o(h^2)$, while $S_{n,2}/S_{n,0} = h^2\mu_2 + o(h^2)$. This is [Eq. 43.1.6](#).

Boundary. For $x = a + ch$ the range of integration in [Lemma 43.1.2](#) is $[-c, (b - x)/h]$, which for large n is $[-c, 1]$. Every moment μ_j above is therefore replaced by $\mu_{j,c}$ and the moment matrix by $\mathbf{A}_c = (\mu_{j+k,c})$, still positive definite for every $c \in [0, 1)$ because the monomials remain linearly independent in $L^2(K du)$ on $[-c, 1]$. For $q = 0$ the leading term is $f'(a)hS_{n,1}/(hS_{n,0}) = f'(a)h\mu_{1,c}/\mu_{0,c} + O(h^2)$, and $\mu_{1,c} \neq 0$ whenever $c < 1$, since the truncated kernel is no longer symmetric. For $q = 1$ the identity $\sum_i \ell_i(x)(x_i - x) = 0$ of [Proposition 43.1.1\(a\)](#) holds exactly, *at every point, boundary or not*, so the first-order term is absent and the same computation as in (a) leaves $\frac{1}{2}f''(a)h^2 \mathbf{e}_1^\top \mathbf{A}_c^{-1} (\mu_{2,c}, \mu_{3,c})^\top + o(h^2)$.

Optimal bandwidth. Write the mean squared error as $Ah^4 + B/(nh)$ with $A = \frac{1}{4}\mu_2^2 f''(x)^2$ and $B = \sigma^2 R(K)/g(x)$. Differentiating, $4Ah^3 = B/(nh^2)$, that is $h^5 = B/(4An)$, which is [Eq. 43.1.7](#); at that h the two terms are in the ratio 1 : 4, so the minimum is $5Ah^4 = 5A^{1/5}\{B/(4n)\}^{4/5}$, the stated value. ■

Idea. Local linear beats local constant twice, and for the same reason both times: the weights of a fit of degree $q \geq 1$ reproduce linear functions *exactly*, whatever the design and wherever the target point. The local linear estimator is therefore blind to how the design points are distributed (no term in g'/g) and to the one-sidedness of a window at the boundary (no term of order h). The local constant estimator is not, and neither extra bias is removed by more data, only by a smaller h .

Part (d) is the rate that every nonparametric method in this chapter obeys. A parametric estimator has variance of order n^{-1} and no bias; here only the $O(nh)$ points in the window contribute, so the variance is of order $(nh)^{-1}$ against a bias of order h^2 . Balancing the two forces $h \asymp n^{-1/5}$ and leaves a mean squared error of order $n^{-4/5}$, strictly slower than n^{-1} . The optimal bandwidth shrinks slowly: ten times the data calls for a window only 1.585 times narrower.

Warning. Eq. 43.1.7 is not a recipe. It involves the unknown $f''(x)$, the unknown σ^2 and the design density, and it is a *pointwise* optimum, so the ideal bandwidth differs at every x . Any formula that gives a bandwidth without an estimate of f'' is smuggling one in.

43.1.4 What the numbers look like

Example. (Bias, variance and the optimal bandwidth)

Take $n = 200$ equally spaced design points on $[0, 1]$ (so $g \equiv 1$), $\sigma = 0.25$ and

$$f(x) = 2x + e^{-25(x-0.35)^2} - \frac{3}{4}e^{-50(x-0.80)^2},$$

a straight line with a bump and a narrower dip. At the top of the bump, $x = 0.35$, $f''(x) = -50.06$, and with the Epanechnikov kernel Eq. 43.1.7 gives $h^* = 0.0715$. Averaging 2000 data sets at each of 29 bandwidths (Figure 43.1.2), the bias at h^* is 0.98 times the asymptotic value $\frac{1}{2}\mu_2 f''(x)h^2$ and the variance 1.02 times $\sigma^2 R(K)/(nh)$, while the empirically best bandwidth is 0.0703. For a first-order expansion at $n = 200$ this is closer agreement than one has any right to expect.

Figure 43.1.1 shows fits at three bandwidths. At the left endpoint the two estimators part company. With $h = 0.08$, the local constant estimator has bias 0.0938 there and the local linear estimator -0.0058 , sixteen times smaller. Halving the bandwidth divides the first by 2.10 and the second by 4.38: order h against order h^2 , as Theorem 43.1.3(c) says.

```
import numpy as np

A1, C1, M1 = 1.0, 25.0, 0.35          # the bump
A2, C2, M2 = -0.75, 50.0, 0.80       # the dip

def f_true(x):
    """The mean function: a straight line plus a bump and a narrower dip."""
    return 2 * x + A1 * np.exp(-C1 * (x - M1) ** 2) + A2 * np.exp(-C2 * (x - M2) ** 2)
```

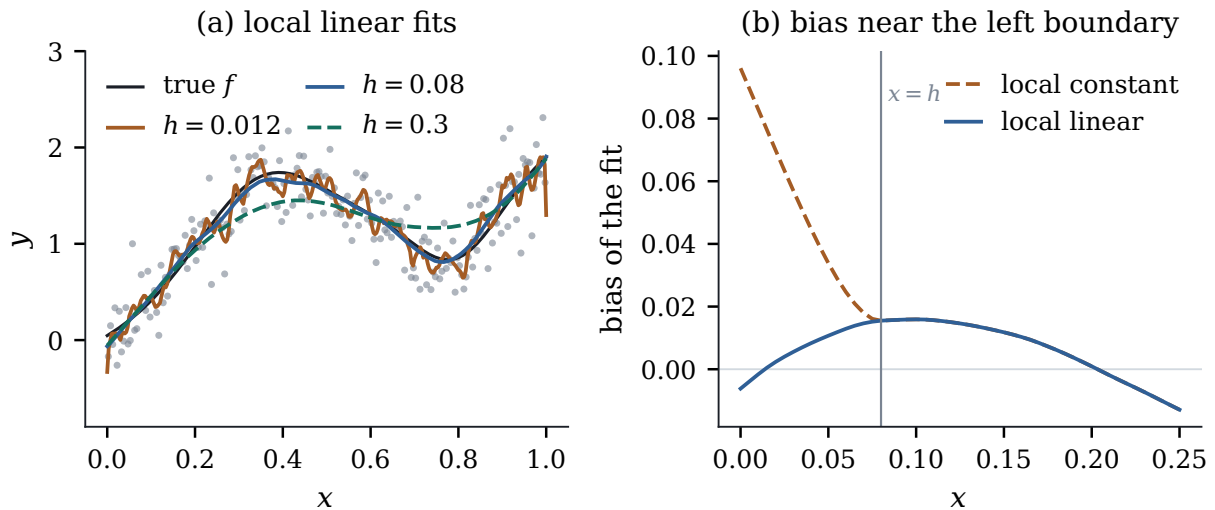


Figure 43.1.1: (a) Local linear fits to one data set at three bandwidths: too small, about right, too large. (b) Bias near the left endpoint over 2000 replicates. Inside $[0, h]$ the local constant window is one-sided, with bias of order h ; the local linear bias stays of order h^2 (Theorem 43.1.3).

```
def epanechnikov(u):
    """K(u) = (3/4)(1 - u^2) on [-1, 1], zero outside."""
    return np.where(np.abs(u) < 1, 0.75 * (1 - u**2), 0.0)

def local_poly_weights(x0, x, h, degree):
    """The weights l_i(x0) of the local polynomial fit: f-hat(x0) = sum_i l_i(x0) y_i."""
    w = epanechnikov((x - x0) / h) / h # kernel weights K_h(x_i - x0)
    U = np.vander(x - x0, degree + 1, increasing=True) # columns (x_i - x0)^j
    A = U.T @ (w[:, None] * U) # the weighted moment matrix
    e1 = np.zeros(degree + 1)
    e1[0] = 1.0
    return np.linalg.solve(A, e1) @ U.T * w # row vector of weights

def local_poly(grid, x, y, h, degree):
    """Local polynomial estimate of degree `degree` on a grid of target points."""
    return np.array([local_poly_weights(x0, x, h, degree) @ y for x0 in grid])

n, sigma = 200, 0.25
x = (np.arange(1, n + 1) - 0.5) / n # the regular design of assumption (A)
rng = np.random.default_rng(20250921)
y = f_true(x) + sigma * rng.normal(size=n)

def endpoint_bias(h, degree):
    w = local_poly_weights(0.0, x, h, degree)
    return w @ f_true(x) - f_true(0.0)

print("bias at the left endpoint, local constant:", round(endpoint_bias(0.08, 0), 4))
```

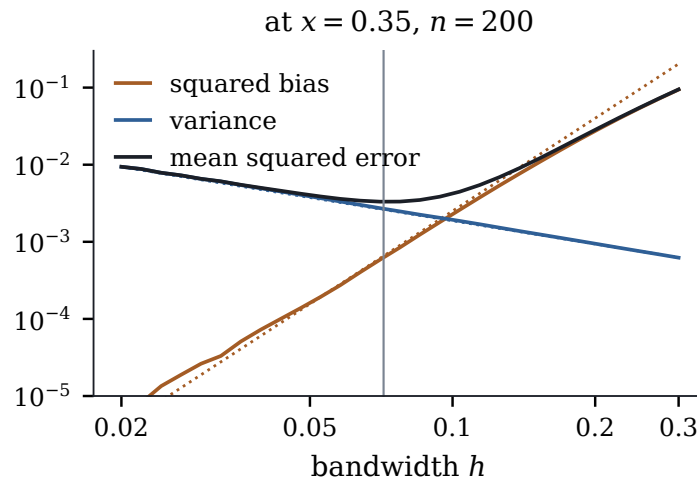


Figure 43.1.2: Squared bias, variance and mean squared error of the local linear estimate at $x = 0.35$ against the bandwidth, from 2000 data sets (solid) and from Theorem 43.1.3(a) (dotted); the vertical line is h^* . Logarithmic scales.

```
print("bias at the left endpoint, local linear: ", round(endpoint_bias(0.08, 1), 4)
)
```

[Run this code online](#)

43.1.5 Choosing the bandwidth in practice, and what kernels cannot do

Two families of rules are used. A **plug-in** rule estimates the unknowns in Eq. 43.1.7 — usually an integrated version of it, with $f''(x)^2$ replaced by $\int f''(t)^2 g(t) dt$ — from a pilot fit, and iterates; Ruppert, Sheather and Wand (1995) is the standard version. A **cross-validatory** rule estimates prediction error directly, and since the local polynomial estimator is linear in $\mathbf{y}$, everything in Section 43.5 applies to it with $\text{tr}(\mathbf{S}_h)$ for $\text{tr}(\mathbf{S}_\lambda)$.

Remark (Loess). The most widely used kernel smoother in practice is Cleveland's **loess** (Cleveland 1979), which makes two changes. The bandwidth is a *nearest-neighbour* one: $h(x)$ is the distance from x to its $\lceil \alpha n \rceil$ th nearest design point, so the window holds a fixed fraction of the data and widens where the design is sparse. And the fit is iterated with bisquare weights on the residuals, the reweighting of Proposition 20.6.1, which makes it resistant to a few wild responses. Neither change affects Theorem 43.1.3 at an interior point where g is continuous, since $h(x) \approx \alpha / \{2g(x)\}$ is then a fixed bandwidth divided by $g(x)$.

There are two reasons this chapter does not stop here. The first is that a kernel estimate is the solution of no global optimization problem: there is no criterion whose minimizer it is and no coefficient vector to report, so there is no obvious way to add a second covariate, a nonnormal response or a random effect. The second is dimension. With d covariates the window is a d -dimensional ball holding of order nh^d points, and the optimal rate degrades to $n^{-4/(4+d)}$ — the *curse of dimensionality*. At $d = 5$ it is $n^{-4/9}$, so matching what 100 observations give in one dimension needs of the order of 10^5 of them; Chapter 44 escapes by assuming additivity, not by smoothing harder. The rest of this chapter writes the estimate as the minimizer of a

penalized criterion, which keeps a coefficient vector, a linear operator and a criterion, all of which generalize.

43.1.6 Exercises

43.1.6.1 A. Check your understanding

Exercise A1.

Show directly from [Eq. 43.1.3](#) that the Nadaraya–Watson weights are nonnegative and sum to one, and deduce that $\min_i y_i \leq \hat{f}(x) \leq \max_i y_i$. Show by example that the local linear estimator has no such property.

Solution. The weights $K_h(x_i - x) / \sum_j K_h(x_j - x)$ are nonnegative because $K \geq 0$ and sum to one by construction, so $\hat{f}(x)$ is a convex combination of the y_i . For the local linear estimator take $n = 2$, $x_1 = 0$, $x_2 = 1$, both inside the window at $x = 2$ with equal kernel weights: the line through the two points is extrapolated, giving $\hat{f}(2) = 2y_2 - y_1$, outside $[y_1, y_2]$ when $y_2 > y_1$.

43.1.6.2 B. Practice

Exercise B1.

Compute μ_2 and $R(K)$ for the uniform kernel $K(u) = \frac{1}{2}$ on $[-1, 1]$ and for the Epanechnikov kernel, and verify the values $\mu_2 = 1/5$, $R(K) = 3/5$ quoted after [Definition 43.1.1](#).

Exercise B2.

By [Theorem 43.1.3\(d\)](#), the smallest achievable mean squared error is proportional to $\{\mu_2^2 R(K)^4\}^{1/5}$. Compute this quantity for the uniform, Epanechnikov and normal kernels (for the normal, $\mu_2 = 1$ and $R(K) = 1/(2\sqrt{\pi})$), and express each as a ratio to the Epanechnikov value. Conclude that the choice of kernel changes the attainable error by a few per cent, whereas [Eq. 43.1.7](#) shows that getting h wrong by a factor of two can change it by much more.

Solution. Write $C(K) = \{\mu_2^2 R(K)^4\}^{1/5}$. Epanechnikov: $\{(1/5)^2 (3/5)^4\}^{1/5} = (0.04 \times 0.1296)^{1/5} = 0.3491$. Uniform: $\{(1/3)^2 (1/2)^4\}^{1/5} = (0.11111 \times 0.0625)^{1/5} = 0.3701$, a ratio of 1.0602. Normal: $R(K) = 0.2821$, so $(0.2821^4)^{1/5} = 0.3633$, a ratio of 1.0408. Six per cent and four per cent. By contrast, replacing h^* by $2h^*$ in $Ah^4 + B/(nh)$ multiplies the first term by 16 and divides the second by 2, and with the terms in the ratio 1 : 4 at the optimum this raises the criterion by a factor $(16 + 2)/5 = 3.6$.

Exercise B3.

Suppose f is exactly linear, $f(x) = \alpha + \beta x$. Show that the local linear estimator is unbiased at every x and for every bandwidth, and that the Nadaraya–Watson estimator is unbiased at x only if $\sum_i K_h(x_i - x)(x_i - x) = 0$. Relate the second condition to

Eq. 43.1.6.

43.1.6.3 C. Going deeper

Exercise C1.

Suppose f is four times and g three times continuously differentiable, and let K be symmetric with $\mu_2 = 0$ and $\mu_4 \neq 0$ — a *fourth-order* kernel, which must take negative values, and which violates assumption (B).

- (a) Carry Eq. 43.1.8 two terms further, to $S_{n,j} = nh^j \sum_{r=0}^3 h^r g^{(r)}(x) \mu_{j+r}/r! + o(nh^{j+3})$, and redo the local constant ($q = 0$) bias calculation in the proof of Theorem 43.1.3 to get

$$E\{\hat{f}(x)\} - f(x) = \mu_4 h^4 \left\{ \frac{f'g'''}{6g} + \frac{f''g''}{4g} + \frac{f'''g'}{6g} + \frac{f^{(4)}}{24} \right\}(x) + o(h^4).$$

Deduce that the optimal bandwidth is of order $n^{-1/9}$ and the mean squared error of order $n^{-8/9}$.

- (b) Show that for $q = 1$ the limiting moment matrix is $\mathbf{A} = \text{diag}(\mu_0, \mu_2) = \text{diag}(1, 0)$, which is singular, so the expansion in the proof does not apply. Writing $a = S_{n,0}$, $b = h^{-1}S_{n,1}$, $c = h^{-2}S_{n,2}$ and $u_i = (x_i - x)/h$, show from Eq. 43.1.4 that $\ell_i(x) = K_h(x_i - x)(c - bu_i)/(ac - b^2)$, that $b/c = O(h)$, and hence that the local linear weights collapse to the local constant ones.
- (c) Why is a fourth-order kernel rarely worth using? Consider the sign of $\hat{f}$ when every $y_i \geq 0$, the size of the constants, and (b).

Solution.

- (a) Symmetry gives $\mu_1 = \mu_3 = \mu_5 = 0$, and $\mu_2 = 0$ by assumption, so the expansion leaves $S_{n,0} = ng(x)\{1 + O(h^4)\}$ and, to relative order $o(1)$, $S_{n,1} = nh^4\mu_4g'''(x)/6$, $S_{n,2} = nh^4\mu_4g''(x)/2$, $S_{n,3} = nh^4\mu_4g'(x)$ and $S_{n,4} = nh^4\mu_4g(x)$. Substituting these in $\sum_{j=1}^4 f^{(j)}(x)S_{n,j}/(j!S_{n,0})$ gives the display: every term of order h , h^2 and h^3 has vanished. Balancing the squared bias h^8 against the variance $(nh)^{-1}$ gives $h \asymp n^{-1/9}$ and mean squared error of order $n^{-8/9}$.
- (b) $\mathbf{A} = (\mu_{j+k})_{j,k=0}^1$ has entries $\mu_0 = 1$, $\mu_1 = 0$, $\mu_2 = 0$; it is singular, and the step of the proof that inverts it fails. For the finite-sample weights, $\mathbf{U}^\top \mathbf{W} \mathbf{U} = \begin{pmatrix} a & b \\ b & c \end{pmatrix}$ in the notation of the proof, whose inverse has first row $(c, -b)/(ac - b^2)$, and $(\mathbf{U}^\top \mathbf{W})_{ji} = K_h(x_i - x)u_i^j$; multiplying gives the stated $\ell_i(x)$. By (a), $b = nh^3\mu_4g'''/6$ and $c = nh^2\mu_4g''/2$ to leading order, so $b/c = O(h)$ and $\ell_i(x) = K_h(x_i - x)/a + O(h)$, the Nadaraya–Watson weights. In floating point the matrix is worse than singular: it is ill conditioned with an entry of indefinite sign.
- (c) The constant now involves $f^{(4)}$, far harder to estimate than f'' ; the gain over $n^{-4/5}$ shows only at sample sizes where the two rates differ appreciably; negative weights can make $\hat{f}$ negative where the data are not, which is unacceptable for a variance or a probability; and by
- (d) the fit loses the boundary and design-bias advantages that were the reason for $q = 1$.

Exercise C2.

Replace assumption (A) by: $x_1, \dots, x_n$ are independent with density g , continuous and positive at x . Show that $E\{S_{n,j}\} = nh^j\{g(x)\mu_j + O(h)\}$ and $\text{Var}(S_{n,j}) = O(nh^{2j-1})$, so that $(nh^j)^{-1}S_{n,j} \rightarrow g(x)\mu_j$ in probability. Explain why this gives the conditional bias and variance of [Theorem 43.1.3](#) with o_p in place of o , and why the argument needs more care than the fixed-design one when the bias term of order h^2 is required (compare the size of the stochastic error $(nh^3)^{-1/2}$ with h at $h \asymp n^{-1/5}$).

Solution. $E\{S_{n,j}\} = n E\{K_h(x_1 - x)(x_1 - x)^j\} = nh^j \int K(u)u^j g(x+hu) du = nh^j\{g(x)\mu_j + O(h)\}$ by continuity of g , and $\text{Var}(S_{n,j}) \leq n E\{K_h(x_1 - x)^2(x_1 - x)^{2j}\} = O(nh^{2j-1})$. The standard deviation of $(nh^j)^{-1}S_{n,j}$ is therefore $O\{(nh)^{-1/2}\}$, which tends to zero under (D), and Slutsky's theorem transfers every limit in the proof. The difficulty is that the stochastic error in the bias is of order $h^2(nh)^{-1/2}$, which at $h \asymp n^{-1/5}$ is $n^{-4/5}$, the same order as the squared bias. A statement about the *conditional* bias given the design therefore needs the sharper expansions of Fan and Gijbels (1996, ch. 3).